Module-03 **RBT Level-**L1, L2

Power Pivot& Dax- Introduction to power Pivot,Importing Data from Multiple Sources (Excel, CSV, etc.), Data Model, Relationships /Data Modeling/ Cardinality, DAX: Enhancing Data model, DAX operators, DAX functions, Text/String function, DAX Statistical Functions, Filter Functions.

Introduction to Power Pivot

Power Pivot is a data modeling technology that lets you create data models, establish relationships, and create calculations. With Power Pivot we can work with large data sets, build extensive relationships, and create complex (or simple) calculations, all in a high-performance environment, and all within the familiar experience of Excel.

Power Pivot: Powerful Data Analysis and Data Modeling in Excel

- PowerPivot is an add-in feature of Microsoft Excel.
- It's a spreadsheet program that extends a local instance of the Microsoft Analysis Services tabular, which is integrated directly into an Excel worksheet.
- This feature allows users to develop a ROLAP model in Power Pivot and then utilize pivot tables to explore the huge data model once it's been done.
- In this sense, it functions as a business intelligence platform that uses professional expression languages to query the model and produce advanced measures.
- The main expression language of PowerPivot is DAX, or Data Analysis Expressions, which allows the user to define measures based on the data model.
- It is a tool that can be useful if the user has to calculate a large amount of data because it can analyze and combine millions of rows of table data in seconds.

Tasks in Power Pivot Tables

- The main purpose of PowerPivot is to provide a method to summarize a large amount of data. And makes it simple to understand by displaying the data properly and easily, so that the user can easily analyze the large numerical data in detail.

- **While Comparing the Sales Data:** Think of a scenario, where a user needs to maintain a large sales data on a monthly basis, which contains several different products. The user needs to analyze, which product is been bringing the most busk. Of course, it can be done manually, but it will take a lot amount of time, to do it one by one. Instead, by using the pivot table, it can be done in just a few seconds.
- **While Creating a List of Employees of Different Departments:** Well power pivot is known for automatically doing the calculations. Think of a case, where, where the user has a list of employees, who belongs to several different departments. and the user needs to sort the data out by the name of each department and the counts of employees this work can be a headache if it has to be done manually. However, by creating the pivot table, the data can be sorted by the name of each department and the names of the employees under the department name. With the help of Excel Pivot, this kind of manual work can be done effectively.
- **In the case shown of Sorting and Combining Duplicate Data:** In this example, think of a scenario, where a user has several duplications in the data, which is affecting the overall sum of data. In this case, the user will have to go through all the data manually to find the duplications and then sort all the data out. Instead, with the help of a pivot table, in just a few steps, the data can be aggregated automatically.
- **It shows Excel the Total Product Sales as Percentages:** usually, pivot automatically shows the totals off each row and column while creating them. But it has some other features that make the work easy. Let's take an example, suppose, a user has quarterly sales data of several different products in Excel sheets, and he turns this data into the pivot table, now the pivot will automatically show the total of each product at the bottom of each column. In addition, if the user needs to know the percentage of each product's sales, it can also be configured in just a few steps.

Top Features of Power Pivot for Excel

- **Data Analysis Expressions (DAX):**The powered pivot uses the data analysis expression or DAX as the expression language. This helps extend the data manipulation capabilities of Excel to allow more sophisticated and complex grouping, calculation, and analysis. Which helps maintain large data sets.
- **Multi-Core Processor:**Another feature of power pivot is the feature of multi-core processing and the memory gigabytes, which can be used for faster calculations and processing of thousands of rows in just some

seconds. Also, with the help of compression algorithms, it is possible to load a large amount of data into the memory in an efficient manner.

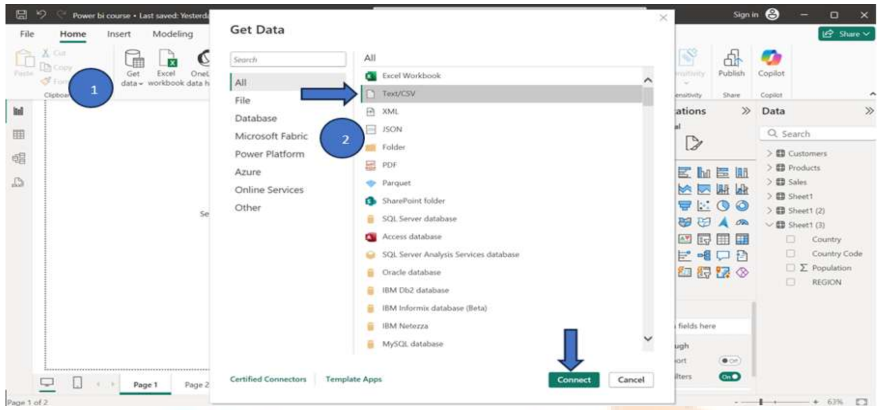
- **Data can be Imported from Multiple Sources:**As previously mentioned, the power pivot allows the user to import and combine data from multiple sources. The data can be sourced from any location for extensive data analysis on the desktop.
- **Save Data Diagram View in the form of Picture:** in the Excel pivot, the data model diagram view can be easily saved as a high-resolution picture, so that it can be later used for printing or analysis. To use this feature, click on the File and select Save the view as the picture.
- **Security and Management Feature:** Another feature of PowerPivot includes security and management, which allows IT managers to measure and control shared applications to assure security, high availability, and performance.
- **Helpful Tool in Editing Process**:With the help of a power pivot, the user can edit the table manually while looking at the sample of data. There can be up to five rows of data in the selected table. With this feature, a user can create and edit data faster and accurately in an efficient manner instead of going back and forth to view data each time.
- **Data Model:**The Data Model is the most powerful feature of Power Pivot. The data that is obtained from various data sources is maintained in Data Model as data tables. You can also create relationships between the data tables so that you can combine the data in the tables for analysis and reporting.
- **Memory Optimization**:Power Pivot Data Model uses xVelocity storage, which is highly compressed when data is loaded into memory which makes it possible to store hundreds of millions of rows in memory.
- **Power Pivot Tables:** Power PivotTables can be created from the Power Pivot window. PivotTable is so created based on the data tables in the Data Model, making it possible to combine data from the related tables for analysis and reporting.
- **Power Pivot Charts**:We can create the Power Pivot Charts from the Power Pivot window. The PivotCharts are created on the basis of data tables in the Data model, making it possible to combine data from the related tables for analysis and reporting.

Applications of Power Pivot.

- **Large-Scale Analysis:** Handles millions of rows efficiently.
- **Data Integration:** Combines data from sources like Excel, SQL Server, or CSV files.
- **Advanced Calculations:** Uses DAX for custom measures and KPIs.
- **Dynamic Reporting:** Creates interactive PivotTables and PivotCharts from related tables.

Importing Data from Multiple Sources

- **Open Power BI Desktop**.
- **Access "Get Data"**: In the **Home** tab of the ribbon, select the "**Get Data**" button.
    - For common sources like Excel or SQL Server, we might see direct icons or a dropdown list.
    - To see the full, extensive list of available sources (files, databases, online services, etc.), select "**More...**".
- **Select Data Source**:
    - In the "Get Data" dialog box, browse or use the search function to find the source you need (e.g., type "CSV" or select "Excel workbook" under the "File" category).
    - Select the source and click "**Connect**".
- **Provide Connection Details**: A connection window specific to chosen source will appear.
    - For local files like Excel or CSV, we will browse to and select the file location.
    - For databases or online services, we may need to enter server names, URLs, and authentication credentials (username and password).
- **Navigate and Preview**: The "**Navigator**" window will display the available data elements (e.g., sheets or tables within an Excel file or database).
    - Check the boxes next to the tables or data want to import. A preview of the selected data will appear on the right.



- **Load or Transform Data**:
  - Select "**Load**" to import the data directly into Power BI model. The data will appear in the "**Data**" pane on the right.
  - Select "**Transform Data**" to open the **Power Query Editor**, which is highly recommended for cleaning, shaping, and combining data before loading it into report.
- **Import Multiple Sources**: Repeat the "Get Data" process for each additional file, database, or online service you need. Each connection will create a separate query in the Power Query Editor.
- **Combine Data (Optional but Recommended)**: In the Power Query Editor, we can combine data from different sources into a unified data model.
  - Use features like **Merge Queries** (to join columns from different tables) or **Append Queries** (to stack rows from tables with a similar structure) from the Home ribbon.
  - Once we satisfied with the combined and transformed data, click "**Close & Apply**" from the Power Query Editor's Home tab to load the final, clean dataset into Power BI Desktop.

Data Modeling in Power BI

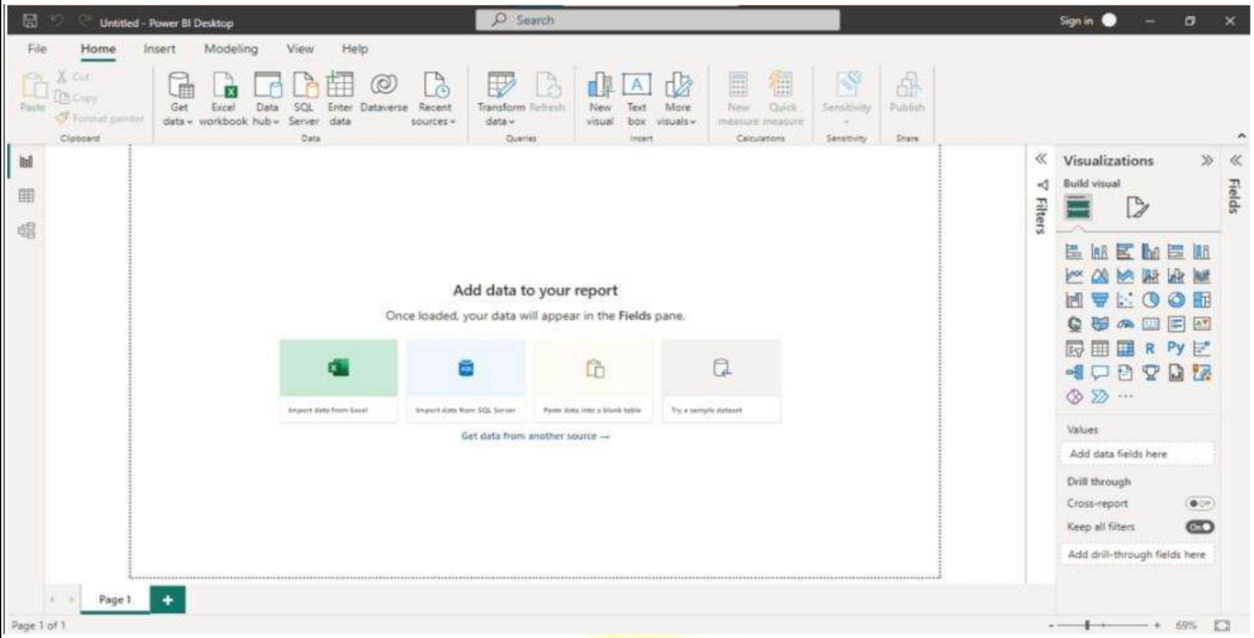
Data modeling is the process of identifying, organizing and defining the types of data a business collects and the relationships between them. It uses diagrams, symbols and textual definitions to visually represent how data is captured, stored and used. A well-designed data model helps:

- Understand data requirements
- Ensure proper structure for reporting
- Align with business goals
- Maintain data integrity

Power BI Interface and Data Loading

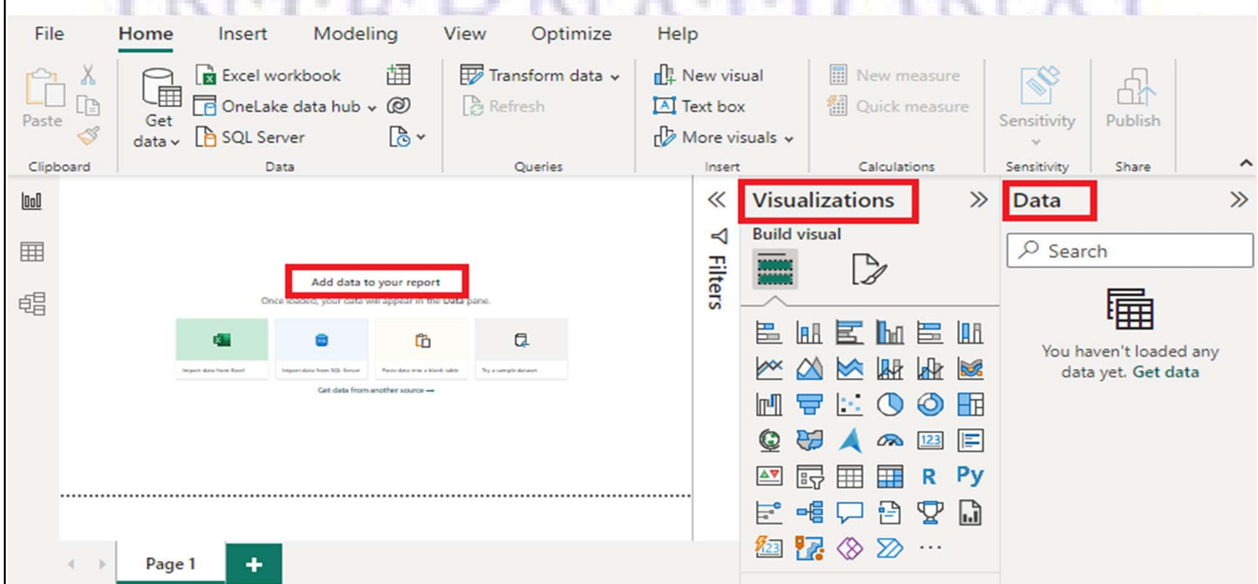
1. Power BI Workspace

Power BI Workspace is the central folder where all datasets, reports, dashboards and workbooks are stored.



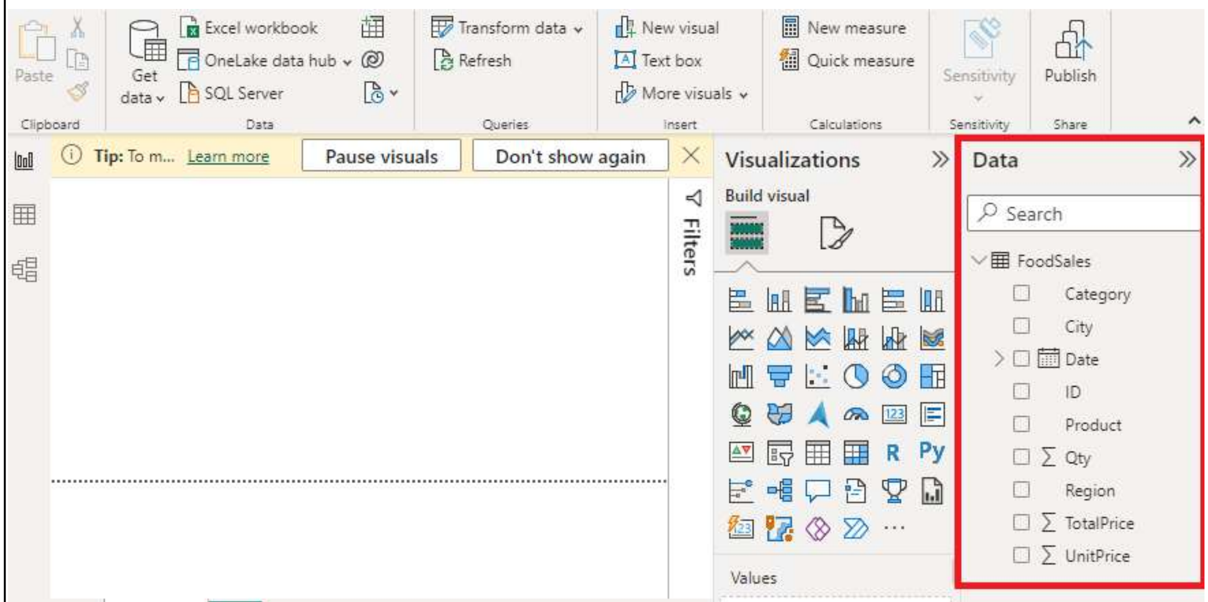
2. Power BI Desktop Panes

Power BI Desktop includes three main panes: Report, Visualizations and Data. The Report pane displays visuals created using fields from the Data pane while the Visualizations pane allows users to select and customize chart types.



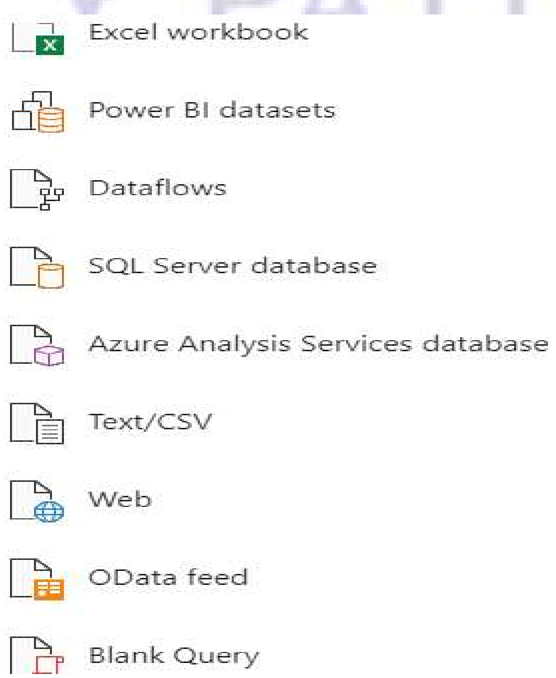
3. Data Pane Overview

After loading a data source such as an Excel file the Data pane on the right side displays tables and fields available for reporting and analysis.



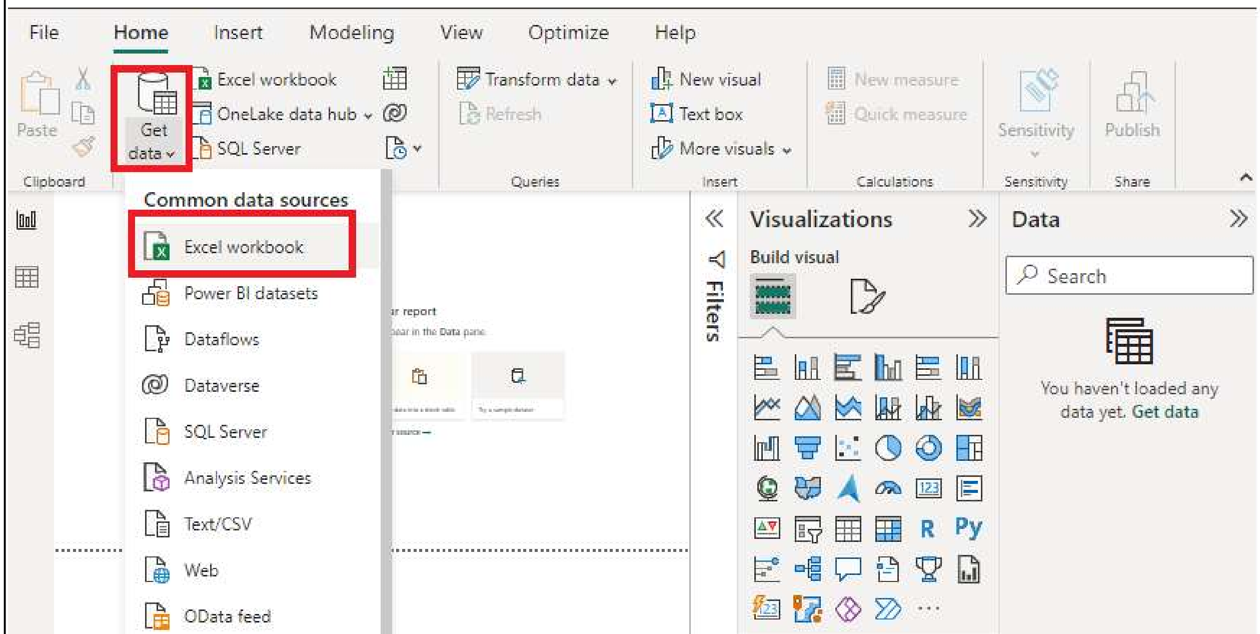
4. Connecting to Data Sources

Power BI allows users to extract data from one or multiple sources. Supported sources include Excel, CSV/Text, XML, JSON, Oracle databases, Azure SQL databases, cloud-based services and more.



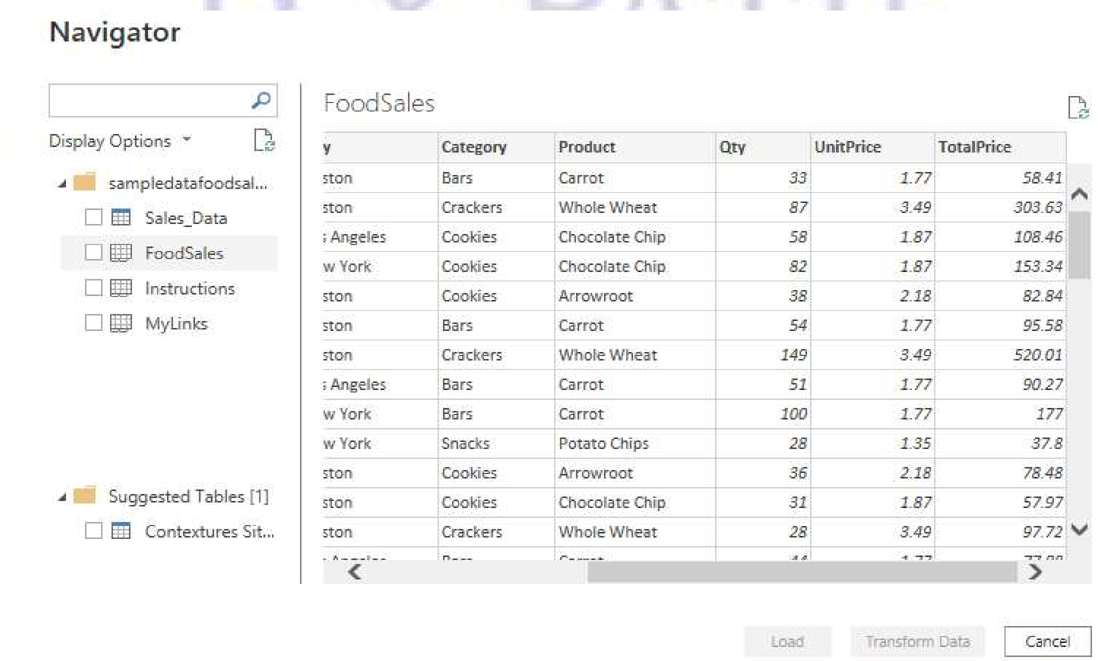
5. Selecting Data Source Type

Users can choose the required data source based on availability and business needs such as Excel files, CSV files, databases or online cloud sources.



6. Data Loading and Transformation

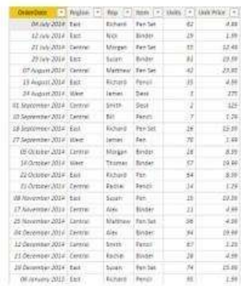
Once a data source is selected the Navigator window helps users preview, transform and load data into Power BI for further modeling and analysis.



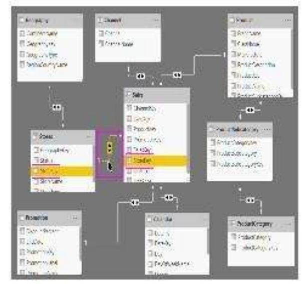
Understanding Relationships in Data Modeling

Relationships are the main feature of data modelling defining all data types. Relationships helps connect with multiple data sources using Cardinality.

(1)

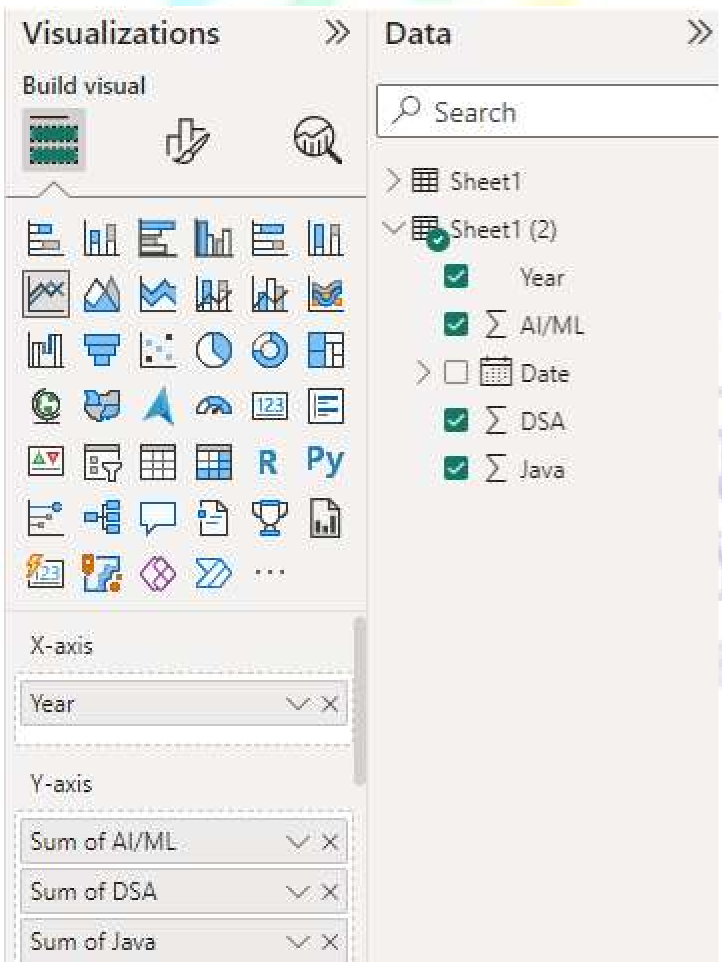


(2)



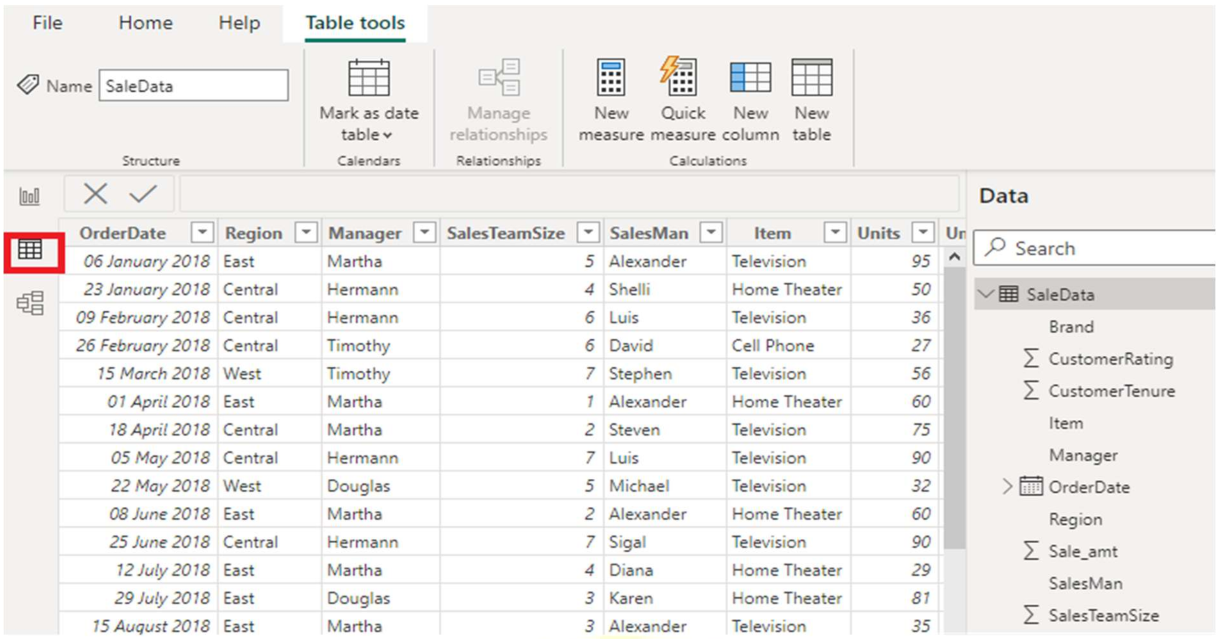
(3)

Data visualizations on multiple data source by analyzing data and defining relationships between them.

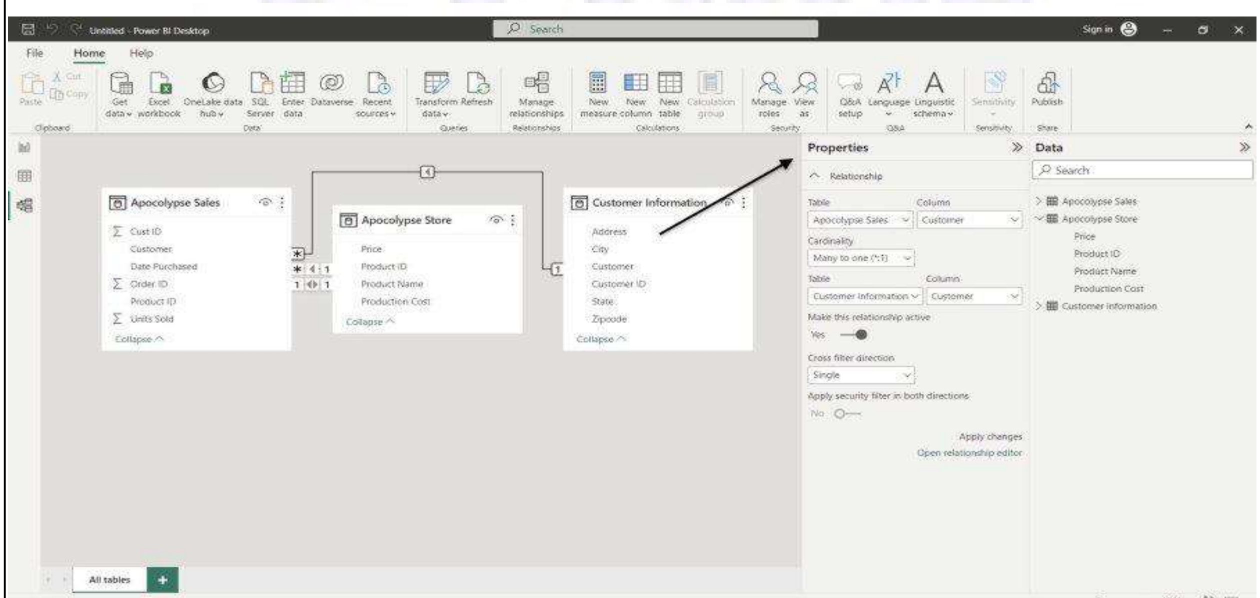


Data view in Power BI

The Data view in Power BI Desktop displays the contents selected data source such as an Excel file. This view allows users to explore tables, columns and individual records, providing a clear understanding of the dataset before creating relationships or reports.

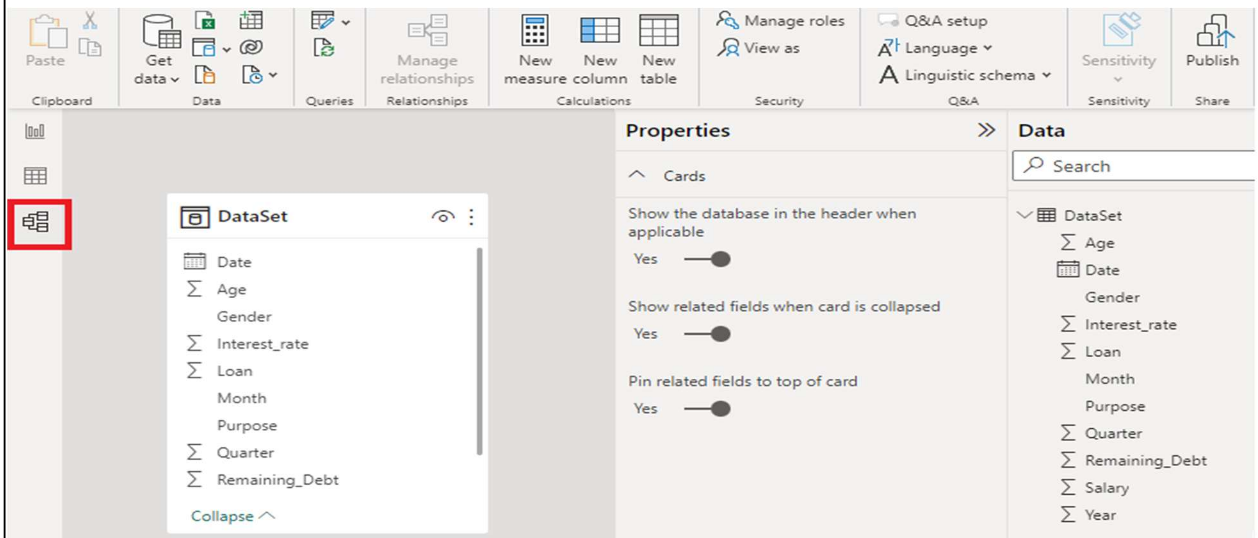


Defining relationships between data attributes in your model enables deeper insights and supports effective data storytelling allowing access to related data across multiple sources



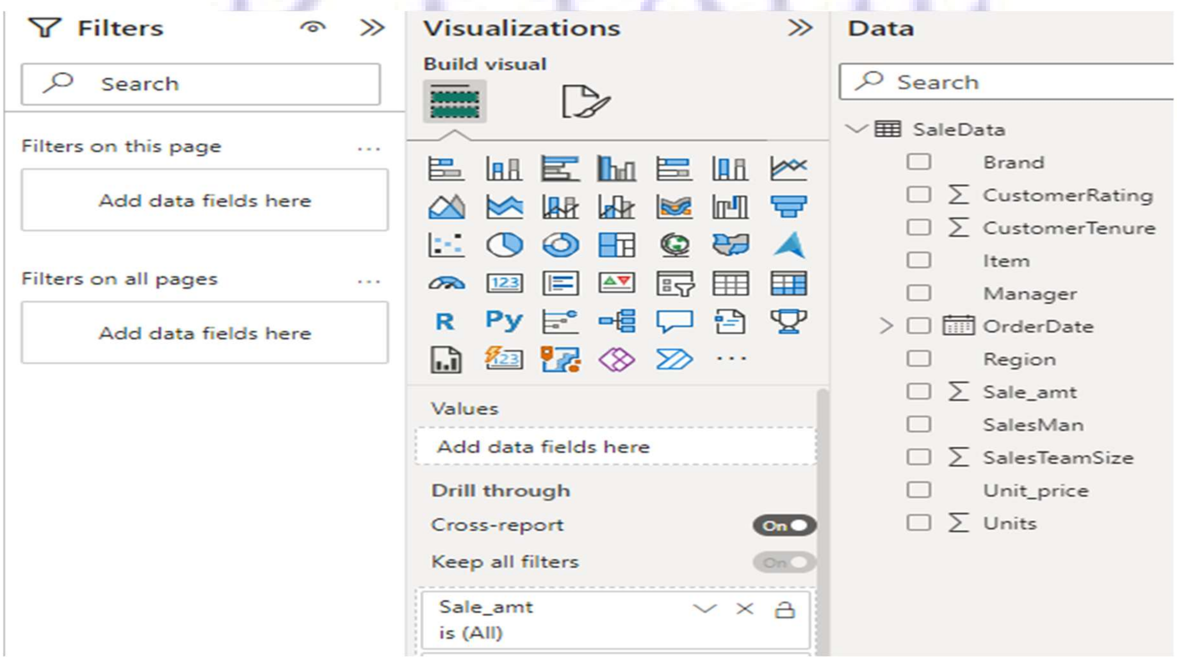
Model View in Power BI

The Model view displays tables, their attributes and the relationships between them which are essential for accurate report generation. The image below shows the Model view for a selected dataset.

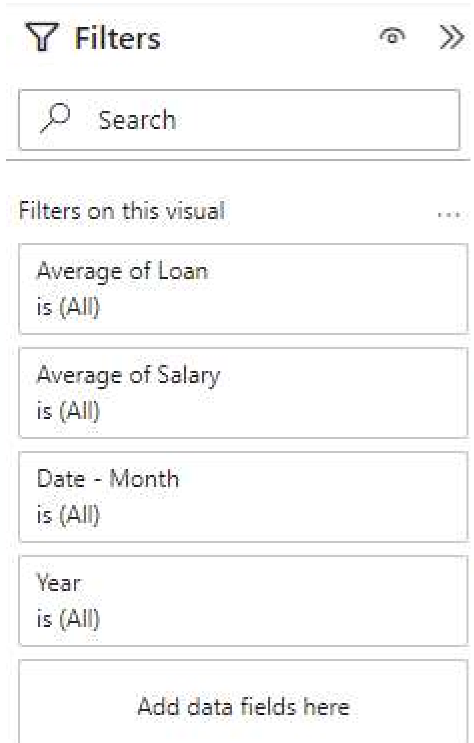


Report View in Power BI

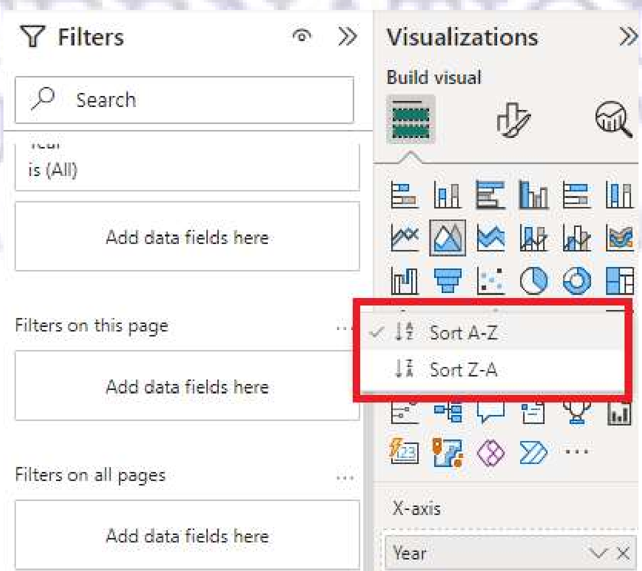
The Report view allows users to create detailed reports with features such as filters, drill-downs and cross-report interactions. Users can select multiple parameters for analysis based on requirements, enabling customized and interactive data exploration.



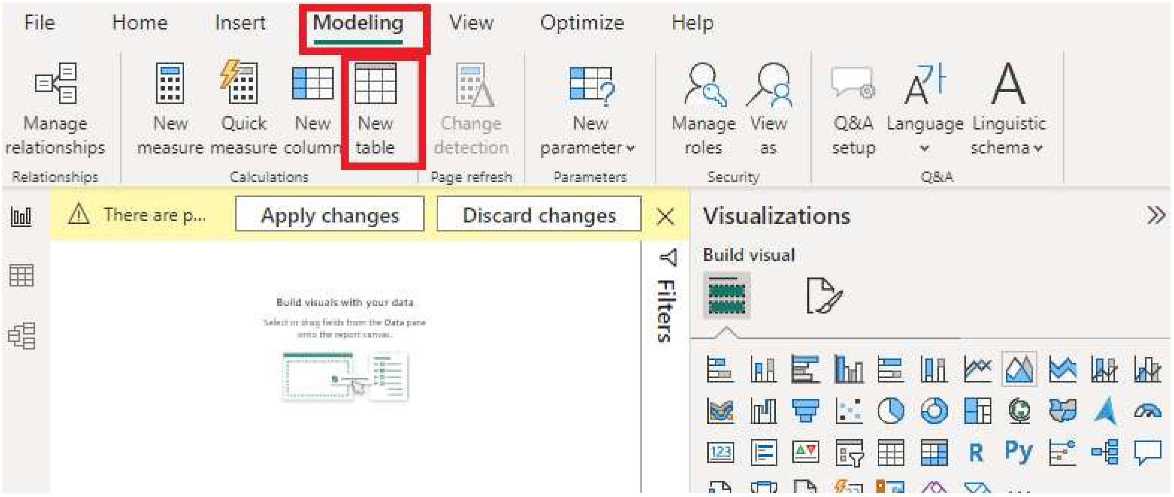
Filter options: Power BI provides various filtering options to refine and customize reports. The image below shows the Filters pane in Power BI Desktop.



Sorting: Users can apply search and sort filters to organize data in reports according to specific requirements.

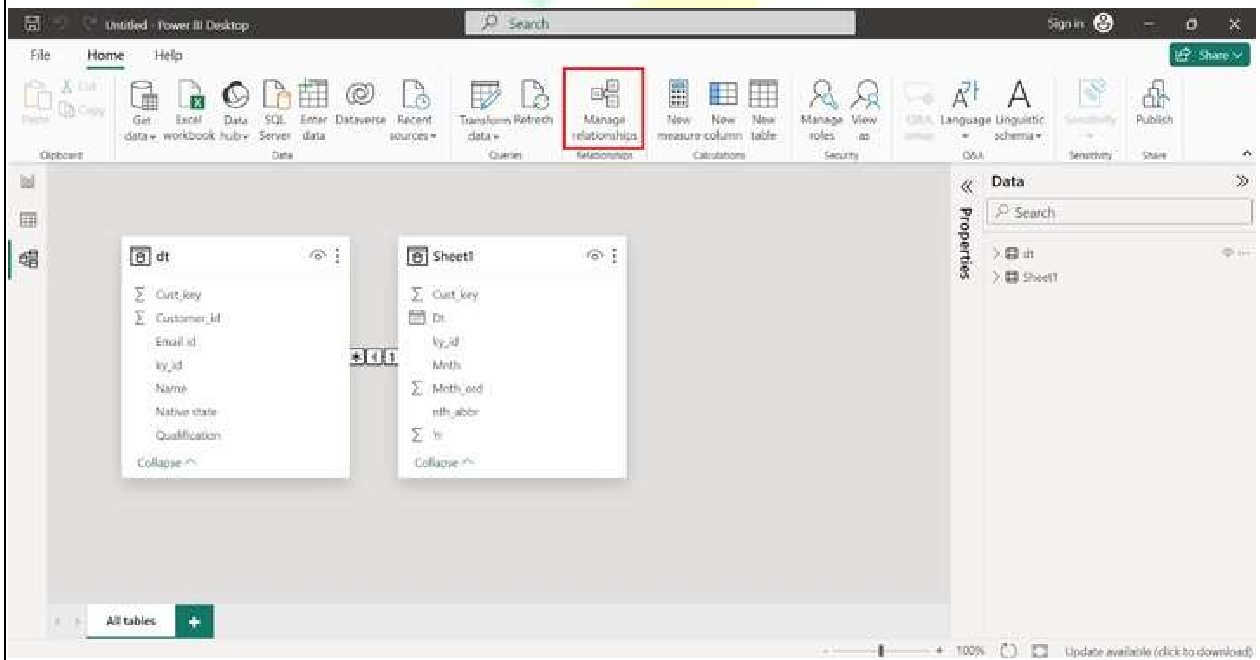


Modelling Feature in Power BI: There is also an option of creating and modelling relationships between table manually.

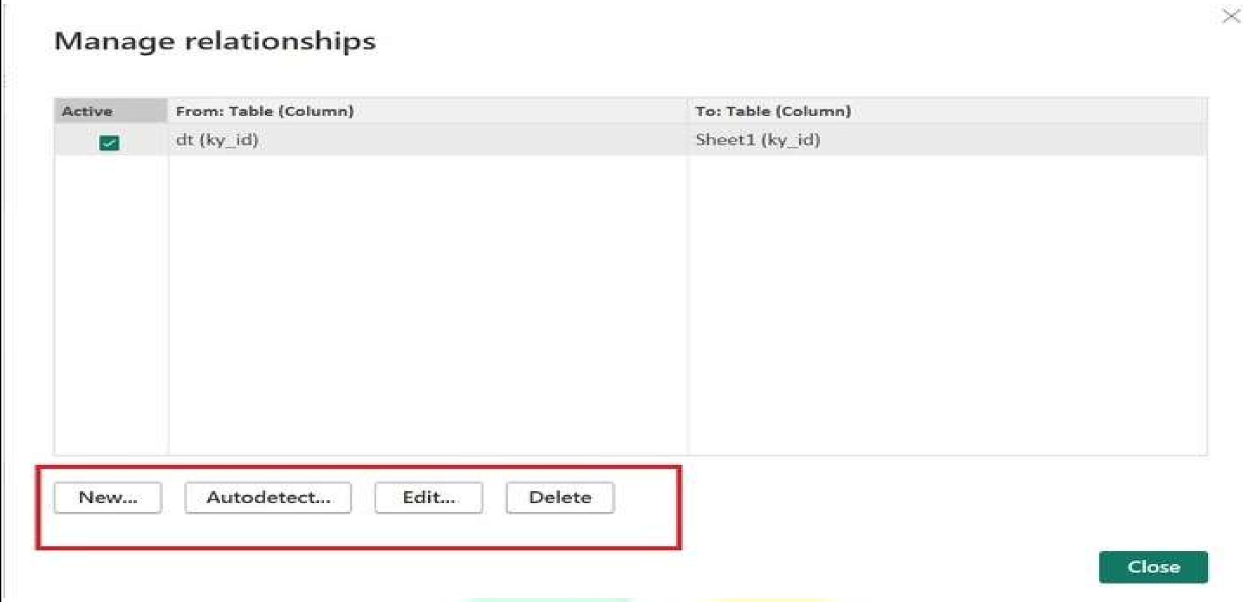


Relationship in Data Modeling

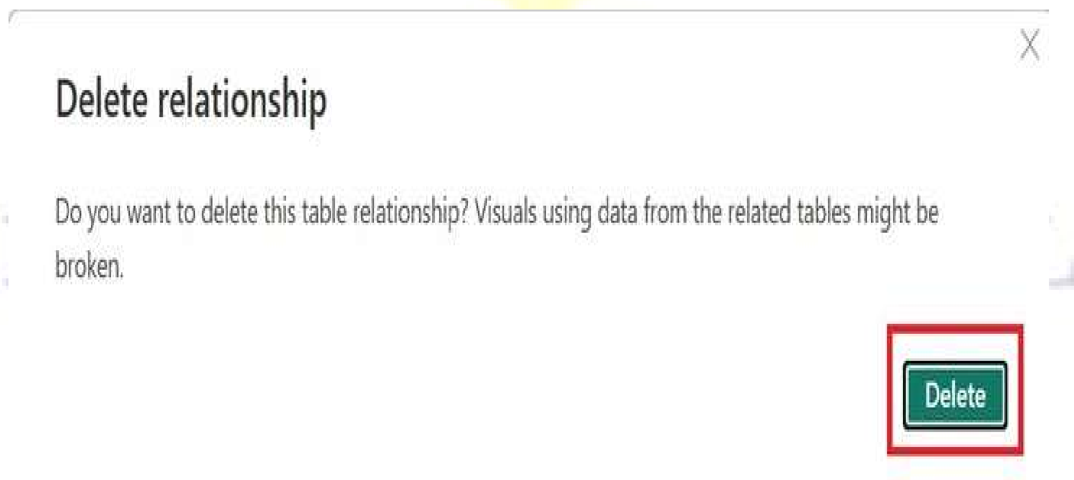
- **Step 1:** We need to navigate to the "**Model View**" and click on the "**Manage Relationship**" under the "**Home**" tab..



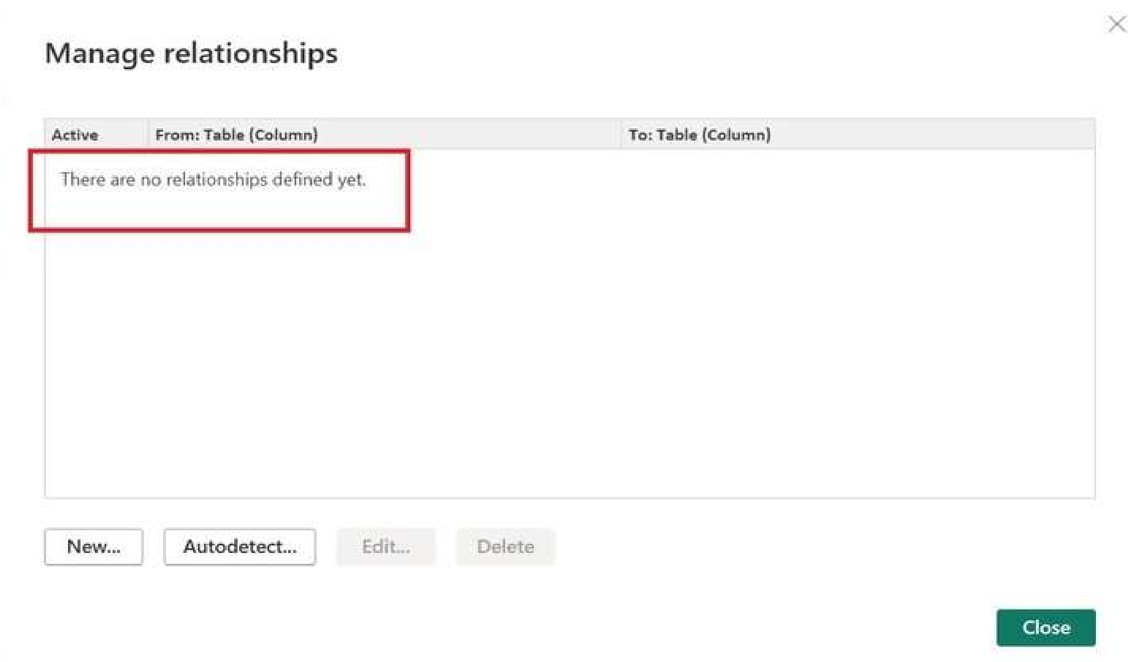
Step 2 : Here is a "**Manage relationships**" dialog box, where we may add the **new relationship**, and **edit** or **delete** the existing relationship. We click on the "**Delete**".



Step 3 : Any window "**Delete relationship**" pops up where We click on "**Delete**".

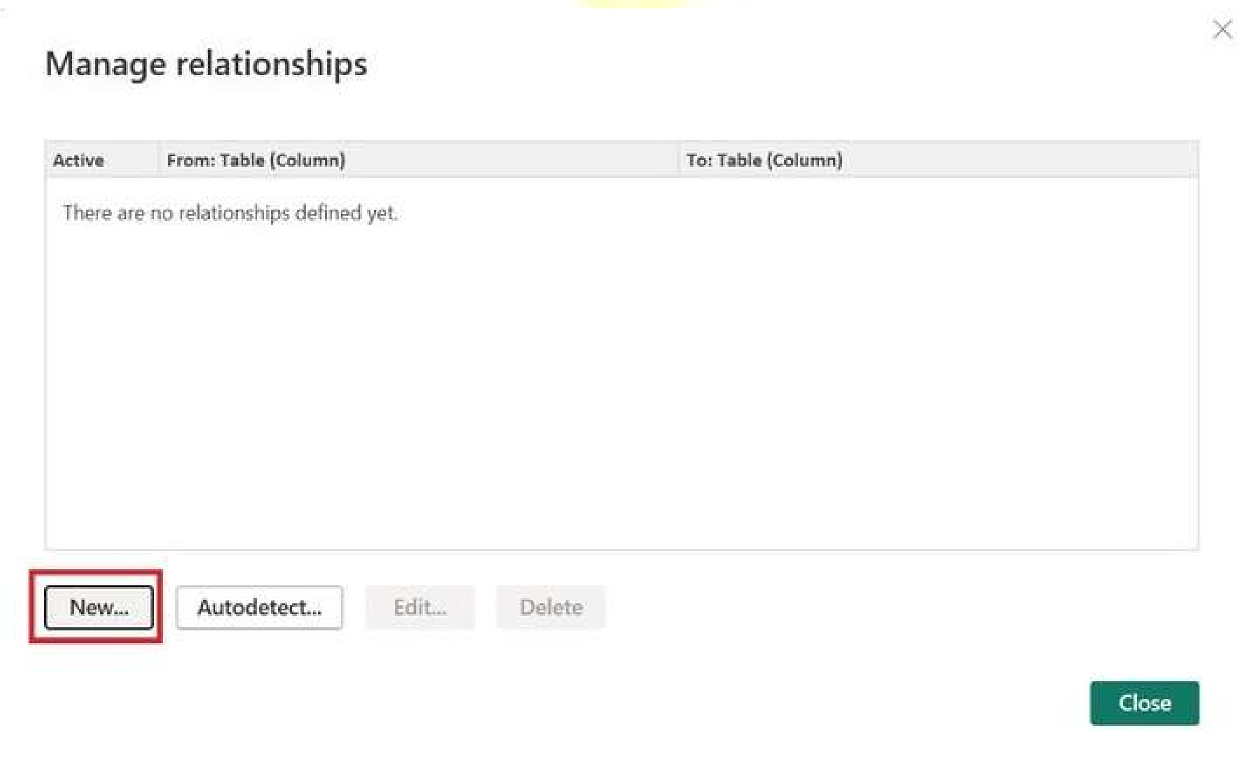


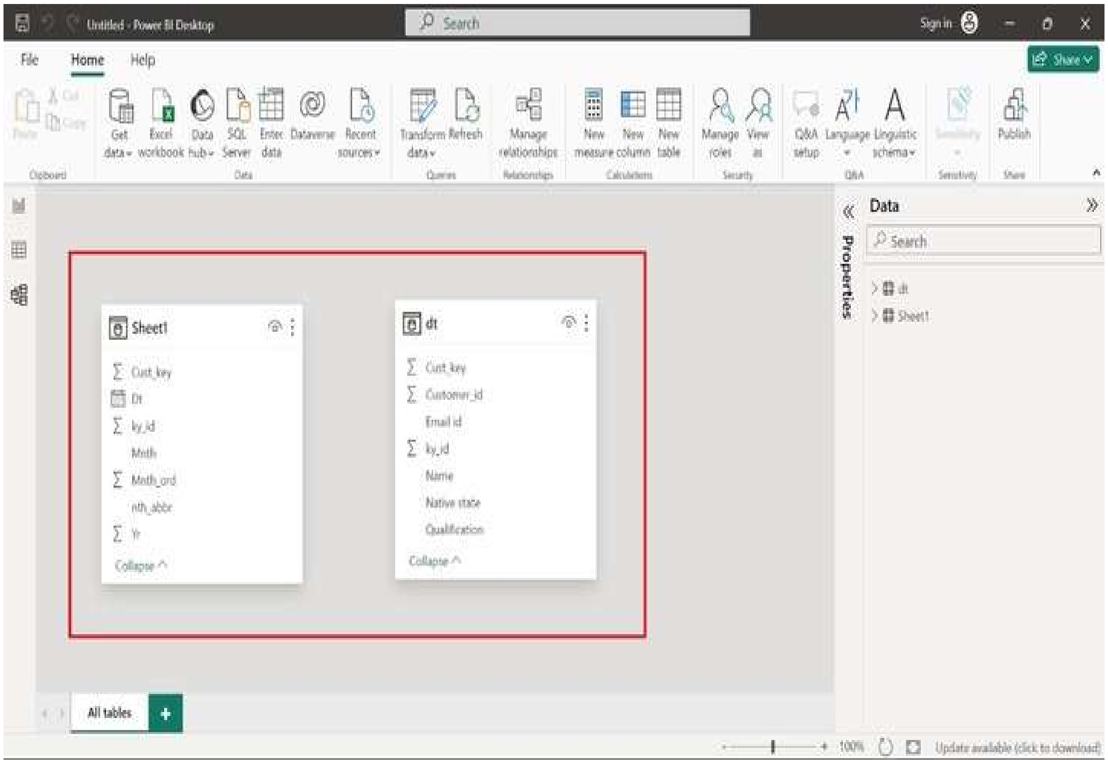
- **Step 4:** As we can see in, the existing relationship between the tables has been deleted.



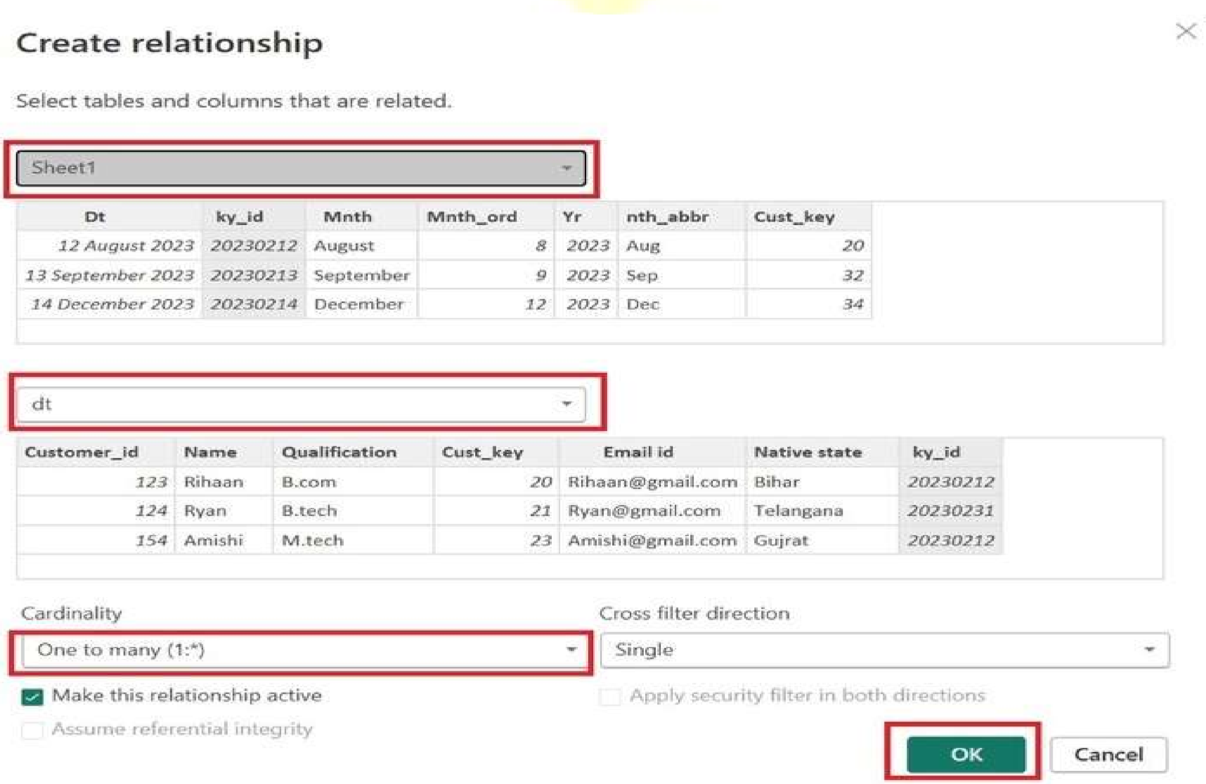
Here, in the **Model View,** there is **no connection** between the "**Sheet1**" and "**dt**" tables.

Step 5:Moreover, We click on the "**New**" tab to create the **new relationship**.

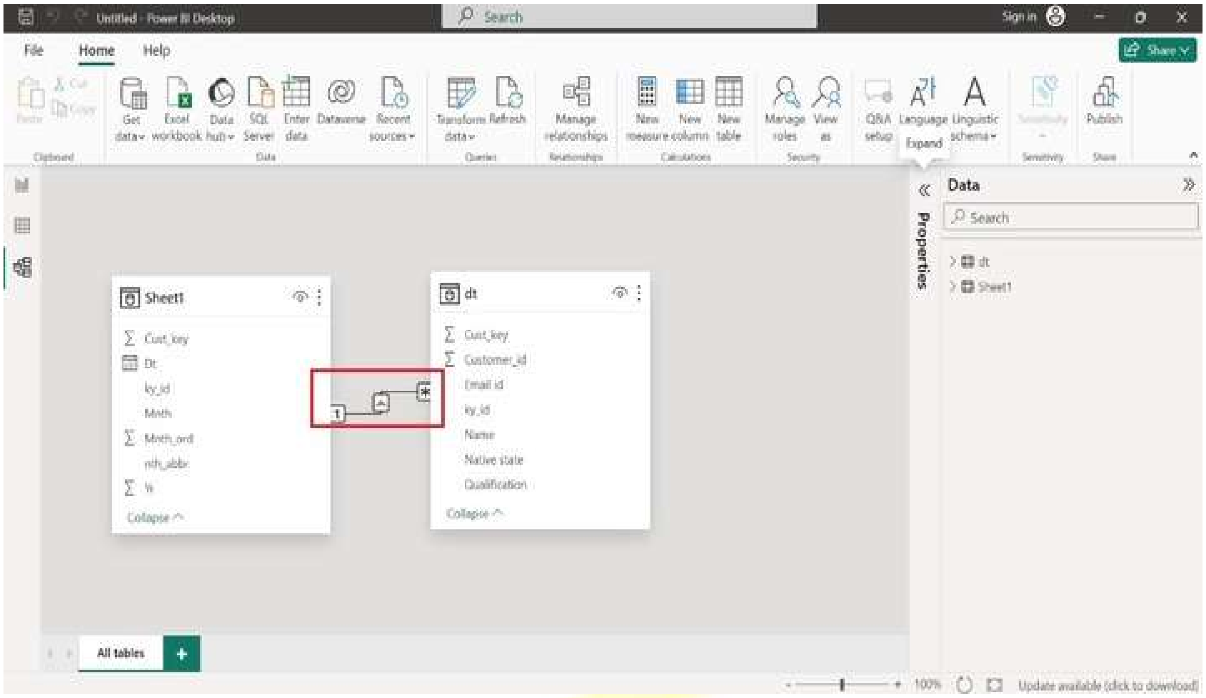




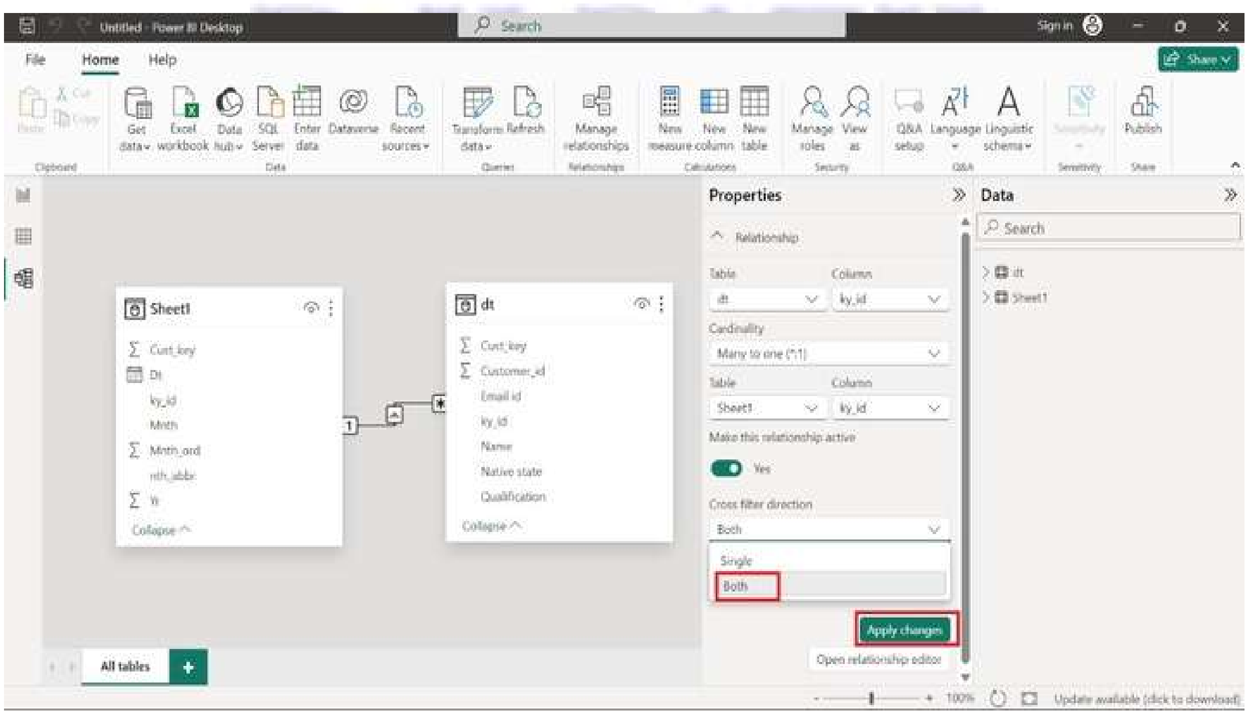
❖ We select "**Sheet1**" as the first table, choose "**dt**" as the second table, and choose "**One to many(1:*)**" under "**Cardinality**" click on **OK.**



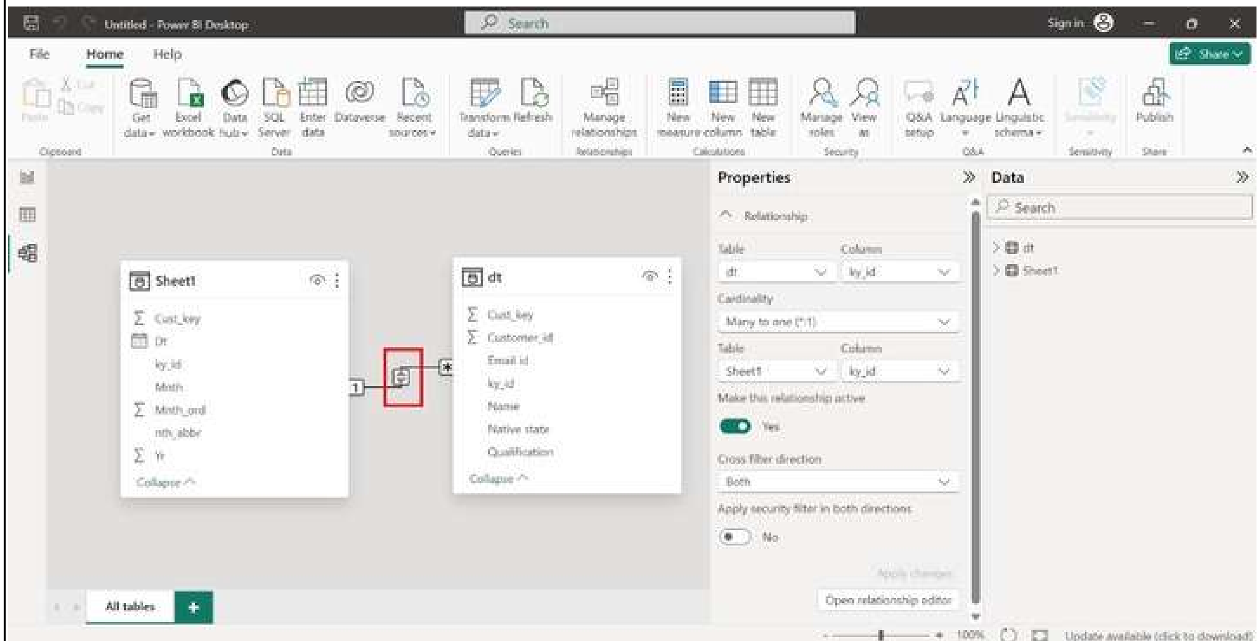
1 means only **one field value** is connected to **multiple filed values (*)** of the common column**(ky_id). The line arrow goes from Sheet1 to dt table which means the Cross-filter direction is single.**



❖ We select "**Both**" under "**Cross filter direction**" from the "**Properties**" pane and click on "**Apply changes**".

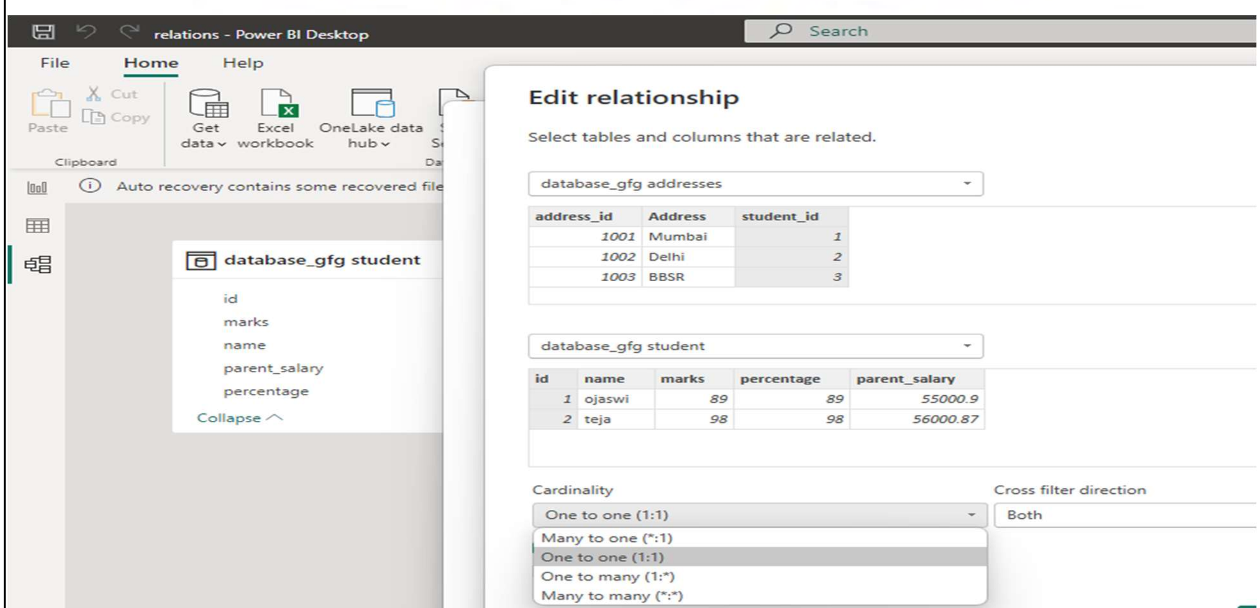


We observe that the **cross-filter direction** has been **modified** and became **bidirectional.**



Cardinalities in BI Desktop

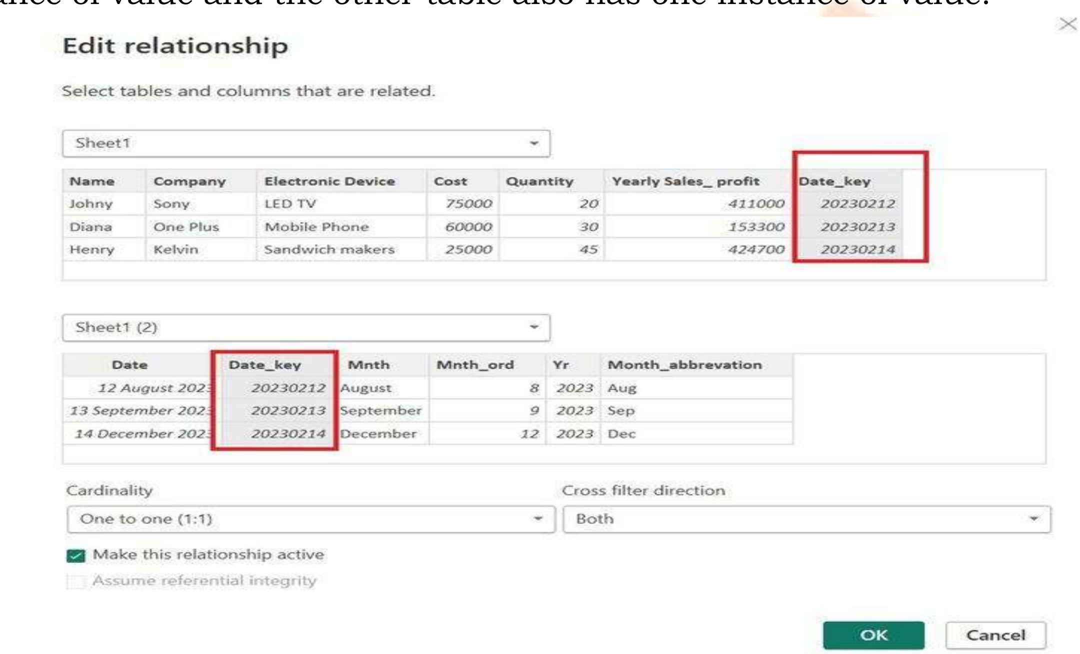
Cardinality describes **how rows in one table relate to rows in another table**. Managing and editing table relationships refers to the process of defining and maintaining the connections between tables in a relational database. Table relationships are important for maintaining data integrity and ensuring efficient data retrieval.



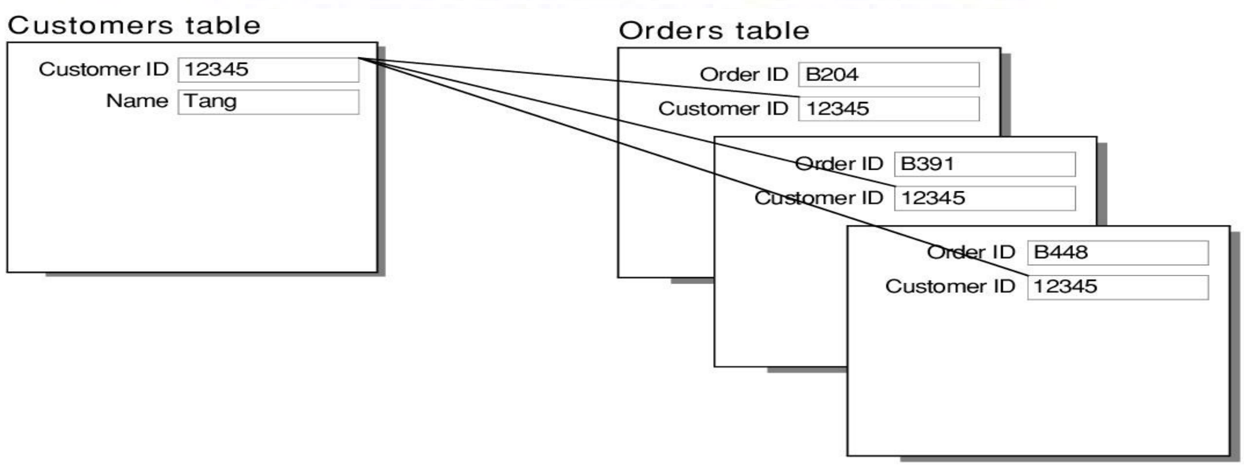
Types of Cardinality in Power BI

- **One-to-One (1:1)**
- **One-to-Many (1:*)**
- **Many-to-One (*:1)**
- **Many-to-Many (:)**

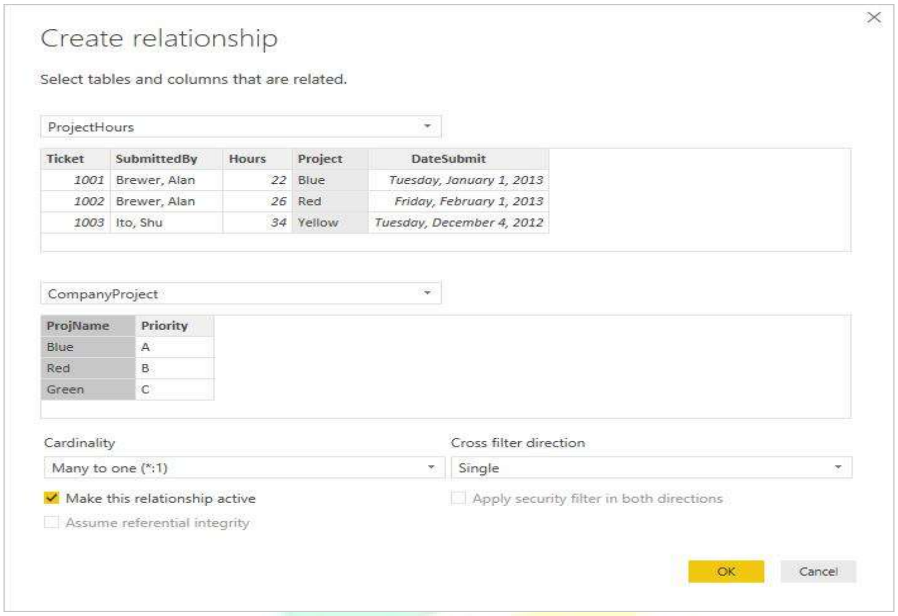
One-to-One (1:1): This means that the column in one table has only one instance of value and the other table also has one instance of value.



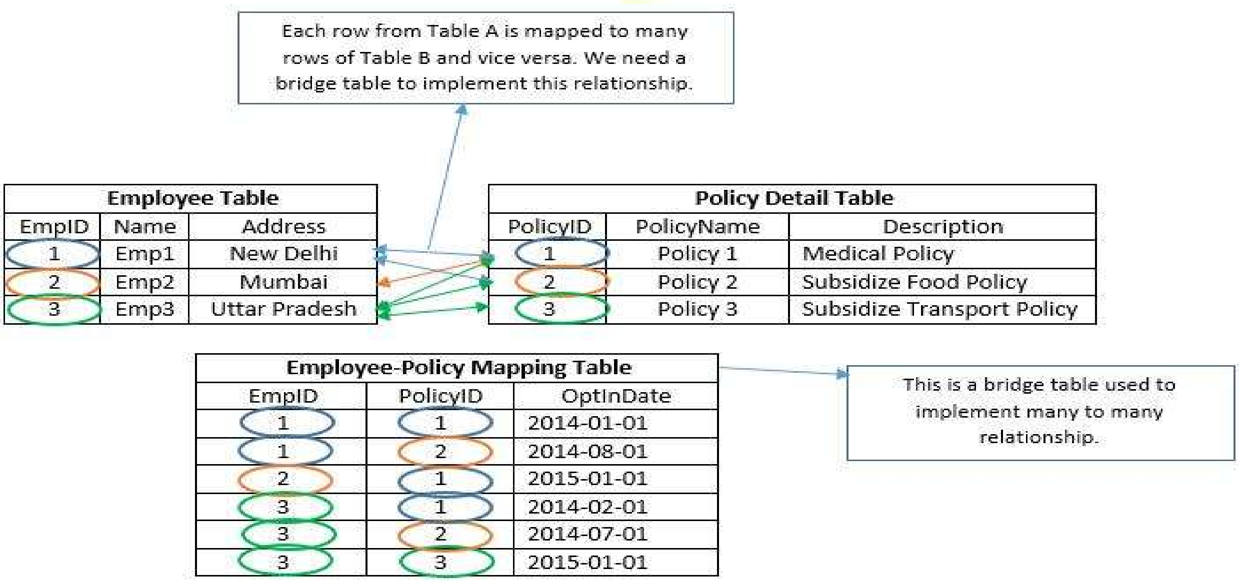
One-to-Many (1:*): In one-to-many relationship, the column in the given table has one instance of value while as the other related table can have more than one instance of value.



Many-to-One (*:1): This means that the column in a given table can have more than one instance of a value while the other table will only have one instance of the value.



Many-to-Many (:): This type of relationship is when a column in both tables has duplicate values.



Data Analysis Expressions (DAX)

Data Analysis Expressions (DAX) is a useful formula and query language used to perform advanced calculations on tabular data models in Power BI and Power Pivot. It enables dynamic, context-aware analysis that goes beyond traditional spreadsheet formulas.

- Used to create custom measures, calculated columns and tables
- Supports complex calculations with row and filter context
- Provides advanced functions not available in standard Excel formulas

DAX uses a formula syntax similar to Excel but extends it with advanced functions designed specifically for tabular data models in Power BI. It allows users to create measures, calculated columns and calculated tables to perform dynamic and context-aware calculations.

Creating Measures for Advanced Calculations

- Measures are a core component of DAX used for calculations on aggregated data
- They are evaluated at query time not stored in the data model
- Measures respond dynamically to filters, slicers and report context
- Commonly used measures include SUM, AVERAGE and COUNT
- DAX supports both implicit and explicit measures
- Using correct data types is essential for accurate measure calculations

Working with Calculated Columns

- Calculated columns create new fields by deriving values from existing columns
- Values are calculated row by row and stored in the data model
- Useful for categorization, flags and static calculations
- Best suited for scenarios where results are not dependent on filters or slicers

Time Intelligence

- DAX includes time intelligence functions for analyzing trends in time-based data
- Functions like TOTALYTD, SAMEPERIODLASTYEAR and DATEADD enable period comparisons

- Used for year-to-date, month-over-month and year-over-year analysis
- Helps generate meaningful insights from time series data

Filtering and Aggregating Data

- DAX provides functions to filter and aggregate data based on specific conditions
- FILTER applies row-level conditions to a table
- CALCULATE modifies filter context to perform dynamic calculations
- SUMMARIZE creates custom aggregations for deeper analysis

Optimization Techniques

- Writing efficient DAX is crucial for fast and responsive Power BI reports
- Use variables (VAR) to simplify calculations and avoid repeated computations
- Minimize complex or nested calculations to reduce processing time
- Understand query behavior and filter context to optimize performance

DAX Operators

- **Arithmetic Operators**
- **Comparison Operators**
- **Logical Operators**
- **Concatenation Operator**

Arithmetic Operators: To perform basic mathematical operations such as addition, subtraction, or multiplication; combine numbers; and produce numeric results, use the following arithmetic operators.

+ (plus sign): Addition

- (minus sign): Subtraction or sign

*** (asterisk): Multiplication**

/ (forward slash): Division

^ (caret): Exponentiation

Comparison operators: We can compare two values with the following operators. When two values are compared by using these operators, the result is a logical value, either TRUE or FALSE.

= (equal to)
> (greater than)
< (less than)
>= (greater than or equal to)
<= (less than or equal to)
<> (not equal to)
== (strict equal to): Differentiates BLANK from zero or an empty string, unlike the standard = operator.

Logical Operators: Use logical operators (&&) and (||) to combine expressions to produce a single result.

&& (Double Ampersand): AND operator. Returns TRUE if both conditions are true (e.g., [Sales] > 100 && [Region] = "USA").
|| (Double Pipe): OR operator. Returns TRUE if at least one condition is true.
NOT (Unary operator): Reverses the logical value
(TRUE to FALSE, FALSE to TRUE).

Concatenation Operator: Use the ampersand (&) to join, or concatenate, two or more text strings to produce a single piece of text.

& (Ampersand): Connects text values (e.g., [FirstName] & " " &[LastName]).

DAX Functions for PowerBI

1. COUNT function in Power BI

COUNT: It returns the total number of items in the column.
Syntax: *COUNT(<column name>)*

COUNTROWS: It returns the number of rows in the table.
Syntax: *COUNTROWS(<table name>)*

DISTINCTCOUNT: It returns the number of distinct values in the values column.
Syntax: *DISTINCTCOUNT(<column name>)*

COUNTA: It counts the number of items in a column, that is not empty i.e. occupied.
Syntax: COUNTA(<column name>)

COUNTBLANK: It counts blank cells in a single column.
Syntax: *COUNTBLANK(<column name>)*

2. DATETIME function in Power BI

DATE: Gets the date in the Date-Time format.
Syntax: *DATE(<year>, <month>, <day>)*

HOUR: Displays hours in the AM or PM format.
Syntax: *HOUR(<datetime>)*

TODAY: It returns today's date.
Syntax: TODAY()

WEEKDAY: It returns values between 1-7 where according to the day number in a week where 1 refers to Sunday and 7 is Saturday.
Syntax: *WEEKDAY(<date>, <return_type>)*

NOW: It returns the current date and time in datetime format.
Syntax: *NOW()*

3. AGGREGATE Functions in Power BI

MIN: Finds the minimum value in a given column or between two expressions.
Syntax: *MIN(<column>) or MIN(<exp1>,<exp2>)*

MAX: Returns the maximum value in a given column or between two expressions.
Syntax: *MAX(<column>) or MAX(<exp1>,<exp2>)*

SUM: The formula adds the values in a column to produce a total.
Syntax: *SUM(Sales[Amount])*

AVERAGE: It takes columns of data and returns the average.
Syntax: *AVERAGE(Sales[Amount])*

MINX: Calculates the minimum value after evaluating each row expression in a table.
Syntax: *MINX(<table>, <expression>)*

4. LOGICAL Functions in Power BI

AND: Combines 2 expressions logically.
Syntax: *AND(<logical1>,<logical2>)*

OR: This function performs the logical disjunction on 2 expressions.
Syntax: *OR(<logical1>,<logical2>)*

NOT: Negates the given expression logically.
Syntax: *NOT(<logic>)*

IF: It checks IF a criterion is true and returns one value if it is and another value if it is not.
Syntax: *IF(<logical_test>,<value_if_true>, value_if_false)*

5. MATH Functions in Power BI
ABS: Returns the absolute value.
Syntax: *ABS([number])*

FACT: Returns the factorial of the number.
Syntax: *FACT([number])*

EXP: Calculates the exponential value of a number.
Syntax: *EXP([Number])*

ROUND: Rounds a number to a specified number of decimal places.
Syntax: *ROUND([Number], [NumDigitsAfterDecimal])*

POWER: Raises a number to a specified power.
Syntax: *POWER([Number], [Power])*

LOG: Calculates the logarithm of a number to a specified base.
Syntax: *LOG([Number], [Base])*

SQRT: Calculates the square root of a number.
Syntax: *SQRT([Number])*

MOD: Returns the remainder of a division operation.
Syntax: *MOD([Dividend], [Divisor])*

SIN, COS, TAN: Calculates the trigonometric sine, cosine and tangent of an angle, respectively.
Syntax: *SIN([Angle]), COS([Angle]), TAN([Angle])*

6. TEXT Functions in Power BI

CONCATENATE: Joins two strings together.
Syntax : *CONCATENATE(<text_1>,<text_2>,...)*

FIXED: Rounds off numbers to a given decimal.
Syntax: *FIXED([Number], [NumDigitsAfterDecimal], [IncludeLeadingZeroes])*

REPLACE: Replace the characters with part of a string.
Syntax: *REPLACE(<Text>, <Old_Text>, <NewText>)*

UPPER: Converts a text string to all uppercase letters.
Syntax: *UPPER(<text>)*

LOWER: Converts a text string to all lowercase letters.
Syntax: LOWER(<text>)

SEARCH: Returns the number of the character at which a searched character or text string is first appeared.
Syntax: *SEARCH(<find_text>, <within_text>[, [<start_num>][, <NotFoundValue>]])*

7. Statistical Functions in Power Bi

MEDIAN: Returns the median value in a column.
Syntax: *MEDIAN([Column])*

MODE: Returns the mode (most frequently occurring value) in a column.
Syntax: *MODE([Column])*

AVERAGE: Calculates the average of a column of numeric values.
Syntax: *AVERAGE([Column])*

STDEV.P: Calculates the population standard deviation of a column.
Syntax: *STDEV.P([Column])*

STDEV.S: Calculates the sample standard deviation of a column.
Syntax: *STDEV.S([Column])*

VAR.P: Calculates the population variance of a column. Example:
Syntax: *VAR.P(Column)*

DAX Filter Functions

The filter functions in DAX are among the most sophisticated and potent and they are very different from Excel operations. The lookup functions operate similarly to a database by employing tables and relationships. You can generate dynamic calculations by manipulating the data context with filtering functions.

- DAX Filter All
- DAX Filter AllCrossFiltered
- DAX Filter AllExcept
- DAX Filter AllNoBlankRow
- DAX Filter AllSelected
- DAX Filter Calculate
- DAX Filter CalculateTable
- DAX Filter Earlier
- DAX Filter Earliest
- DAX Filter FILTER
- DAX Filter KeepFilters
- DAX Filter RemoveFilters

DAX Filter All

It Ignores any filters that may have been used and returns all the rows in a table or all the values in a column. This function can be used to remove filters and do calculations across all table rows.

Syntax: *ALL([<table> | <column>[, <column>[, <column>[,…]]])*
table: The table upon which intend to clear the filters.
column: The column upon which intend to clear filters.

DAX Filter AllCrossFiltered

It Removes all table-related filters that have been active.
Syntax: *ALLCROSSFILTERED(<table>)*
table: The table upon which you intend to clear the filters.

DAX Filter AllExcept

Except for context filters that have been applied to the specified columns, remove all context filters from the table.

Syntax: *ALLEXCEPT(<table>,<column>[,<column>[,…]])*
table: The table, except for context filters on the columns supplied in subsequent parameters, over which no context filters are applied.
column: The column that context filters need to be kept in place for.

DAX Filter AllNoBlankRow

Returns all rows except the blank row or all distinct values of a column except the blank row, from the parent table of a relationship, ignoring any potential context filters.

Syntax: *ALLNOBLANKROW([<table> | <column>[, <column>[, <column>[,…]]])*
table: The table that has had all context filters removed from it.
column: A column that is devoid of all context filters.

DAX Filter AllSelected

Retains all other context filters or explicit filters while removing context filters from columns and rows in the current query. While maintaining explicit filters and contexts other than row and column filters, the ALLSELECTED function obtains the context that represents all rows and columns in the query. Visual totals for queries can be obtained with this function.

Syntax:*ALLSELECTED([<tableName> | <columnName>[,<columnName>[,<columnName>[,…]]])*
tableName: The name of an already-existing table in DAX format. An expression cannot be used for this argument. This element is not required.
columnName: The name of an already-existing table in DAX format. An expression cannot be used for this argument. This element is not required.

DAX Filter Calculate

Interprets an expression in the context of a changed filter.

Syntax: *CALCULATE(<expression>[, <filter1> [, <filter2> [, …]]])*
expression: The expression that must be assessed.
filter1, filter2…: Filter modifier functions, table expressions or Boolean expressions that define filters though optional.

DAX Filter CalculateTable

A table expression is evaluated in a changed filter context.

Syntax: *CALCULATETABLE(<expression>[, <filter1> [, <filter2> [, …]]])*

expression: The expression that must be assessed.

filter1, filter2…: Filter modifier functions, table expressions or Boolean expressions that define filters though optional.

DAX Filter Earlier

In an outer evaluation pass of the given column, returns the current value of the provided column. When you wish to use a certain value as an input and perform calculations depending on that input, EARLIER is helpful for nested calculations. Such calculations are only possible in the context of the current row in Microsoft Excel; but, in DAX, you may store the value of the input and perform the calculation using data from the entire table. EARLIER is frequently used with computed columns.

Syntax: *EARLIER(<column>, <number>)*

column: A column or an expression with a column resolution.

number: A passing score on the external evaluation. One level out from the current evaluation is denoted by 1, two levels out by 2 and so forth. The default value is 1 when omitted. (Optional)

DAX Filter Earliest

It gives back the value of the supplied column as of the most recent outer evaluation pass.

Syntax: EARLIEST(<column>)

column: A mention of a column.

DAX Filter FILTER

A table that displays a subset of another table or expression is returned.

Syntax: FILTER(<table>,<filter>)

table: A table that will be filtered. The table might also be a table-producing expression.

filter: A Boolean expression that must be tested against each table row. Simply a True or False test.

DAX Filter KeepFilters

It alters the way filters are used when a CALCULATE or CALCULATETABLE function is evaluated.

Syntax: KEEPFILTERS(<expression>)

expression: The expression that must be assessed.

DAX Filter RemoveFilters

Remove all filters from the given columns or tables.

Syntax: REMOVEFILTERS(<table> | <column>[, <column>[,…]])

table: The table you wish to clear the filters from.

column: The column on which you want to remove all filtering.

Advantages of DAX

- Enhances data modeling and reporting in Microsoft BI tools such as Power BI and Power Pivot.
- Supports complex formulas, aggregations and conditional logic for deep analytical insights beyond simple arithmetic.
- Works natively across Microsoft's BI ecosystem, ensuring consistent data analysis and modeling.
- Enables creation of custom measures, calculated columns and tables to suit specific business requirements.
- Includes built-in functions for time-based analysis like Year-to-Date (YTD), Quarter-to-Date (QTD) and rolling averages.
- Helps define relationships, hierarchies and calculated entities for accurate, efficient analytical models.
- Supports optimization techniques to enhance query speed and overall responsiveness.
- Backed by a global user base offering tutorials, best practices and learning resources.

S.No	Questions
01	**Define** Power Pivot? **List** and **explain** key features and task Transformed in Power Pivot.
02	**Explain** basic steps required Importing Data from Multiple Sources in Power Pivot.
03	**Define** data modelling and **explain** different **steps** involved in **p**ower BI Interface and Data Loading.
04	**List** and **explain** steps involved in relationship in Data Modeling.
05	**Explain** how to add Power BI Filter to a Page?
06	**Explain** how to add Power BI Filter to a report?
07	**Define** Cardinality? **List** and **explain** different types of Cardinalities.
08	**Define** Data Analysis Expressions (DAX), **explain** how creating Measures for Advanced Calculations.
09	**Explain** how Working with Calculated Columns
10	**Write** short note on 1) Time Intelligence in DAX 2) Filtering and Aggregating Data 3) Optimization Techniques
11	**Explain** DAX Operators.
12	**List** and **explain** COUNT function in Power BI
13	**List** and **explain** DATETIME function in Power BI
14	**List** and **explain** AGGREGATE function in Power BI
15	**List** and **explain** LOGICAL function in Power BI
16	**List** and **explain** MATH function in Power BI
17	**List** and **explain** TEXT function in Power BI
18	**List** and **explain** Statistical function in Power BI
19	**List** and **explain** DAX Filter function in Power BI
20	**Explain** Advantages of DAX